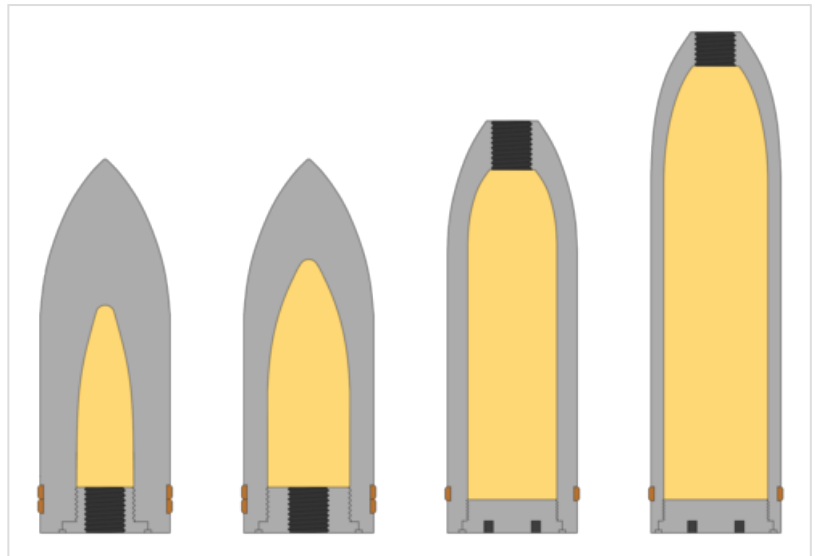




Mine shell

A **mine shell** (from German: *Minengeschoss*, "mine shot") or **high-explosive, high-capacity (HEHC)**^[a] in British military nomenclature,^{[1][2][3]} is a military explosive shell type characterized by thin (usually steel) shell walls which allow a much higher explosive content than standard high-explosive shells of the same caliber, trading a higher pressure wave effect against high-explosive shells' larger fragmentation effect and better penetration of armoured targets.

Mine shells were originally developed during the mid- to late 1800s against fortresses before the introduction of rebar; reinforced fortresses had made the original use of the type obsolete around World War I, but they were given a new role against aircraft during World War II.



Some classic shell types placed from left to right in relation to explosive filler (in yellow).

From left to right:

- APHE – armor-piercing, high-explosive
- SAPHE – semi armor-piercing, high-explosive
- HE – high-explosive
- HEHC – **high-explosive, high-capacity (mine shell)**

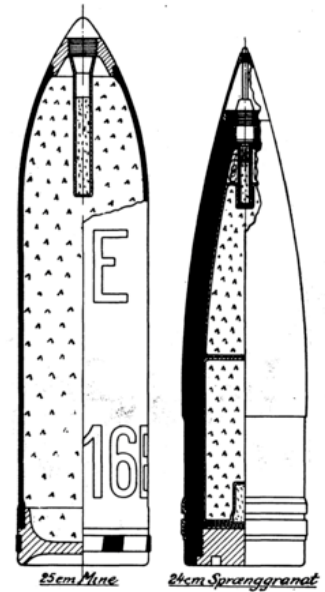
Effect, construction and use

The mine shell is a more explosive version of the common high-explosive and high-explosive fragmentation shells, relying on inflicting damage primarily through the blast (pressure wave) alone rather than the higher fragmentation but less blast provided by standard high-explosive shells. This is referred to as pressure wave effect/damage in, for example, Swedish (Swedish: *tryckvågsverkan*).^[4] This effect is desirable when attacking non-reinforced and non-armoured targets such as brittle brick or concrete, and aircraft skin, as these are relatively easy to penetrate and do not need to be attacked by heavy, hard projectiles, but are tough enough to maintain their structure despite being pierced by shellfire and fragmentation. The larger explosions generated by mine shells are more efficient at inflicting damage on such targets than the greater kinetic impact but smaller detonations delivered by conventional rounds. To achieve this effect mine shells feature very thin shell walls and thus more room for explosive filler, at the cost of generating lighter and thus somewhat less formidable shrapnel (fragmentation).^{[5][4]} Against thicker targets, such as brick or concrete walls, they often also feature a delayed action on the fuze so they can penetrate into the wall and burst inside it, forcing larger pieces of the targets to break loose.

An additional advantage of the mine shell is that, as explosives are less dense than metal, the projectiles weigh less, giving them higher muzzle velocity than heavier shells, and generating less recoil. However less desirably, the reduced mass inevitably entails that they will possess less momentum, which reduces their range as velocity falls off more rapidly than in a heavier projectile with more inertia. It is also a consideration that the lower recoil makes them unsuitable to be fired from the same gun as standard shells of the same caliber if it uses a recoil operated or advanced primer ignition mechanism. Mine shells are consequently often made longer than other shells of the same caliber, both to increase the weight to about the same as standard shells and to further increase explosive energy.



Upper image:
British test of a single German 30 mm mine shell hitting a Bristol Blenheim bomber aircraft.



Right image:
Cut-through drawing of WWI-era shell types. To the left is a 25 cm mine shell; to the right a 24 cm high-explosive shell.

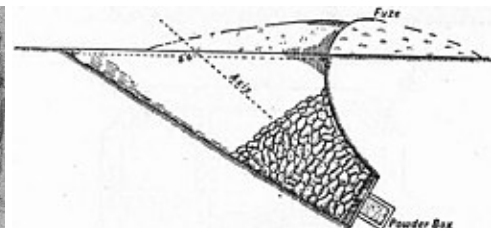
Name

Meaning

The word 'mine' in the name "mine shell" can in the modern world seem confusing, as military use of the word is mainly associated with land mines and naval mines. However the word "mine" is very old and originally had the same meaning as "mining mines".^[6] As mines were used during siege warfare in past eras to collapse fortifications, and later blowing up fortifications (see tunnel warfare), the first proto-landmines developed received the name mine.^[6] These were basic explosives dug down into the ground like a mine. This eventually led to mine-ordnance being defined as "contained explosions", which is alluded to in the name mine shell as its original purpose was to penetrate into fortification walls and burst inside.



Mine-construction during the American Civil War. This mine was dug under the enemy like traditional tunnel warfare. (see Battle of the Crater)



French "fougasse" mine-construction from pre-modern times.^[6] This weapon-name was later given by the French to a specific mine shell design called *obus fougasse* (fougasse shell).^[7]

However, as mine shells became obsolete against fortresses the definition changed with time from "shells damaging through a contained explosion" to "shells damaging through the shock wave created from the explosion of their payload, rather than the combination of fragmentation and pressure wave damage like

traditional high-explosive shells, which have thicker shell walls and smaller explosive load".^{[5][4]}

Spread and use

The name 'mine shell' in English is a modern term directly translated from the German military term *Minengeschoß* ("mine shot").^[8] The historical equivalent English military term is *High-Explosive, High-Capacity*,^{[1][3]} abbreviated as *HEHC*,^[2] which has become uncommon or obsolete in modern times. The name mine shell is or has however been present in several different military nomenclatures around the world besides Germany and the United Kingdom; although in modern times only a few countries still use the term "mine shell" (or equivalent) and its umbrella terms in military nomenclature.

International terms

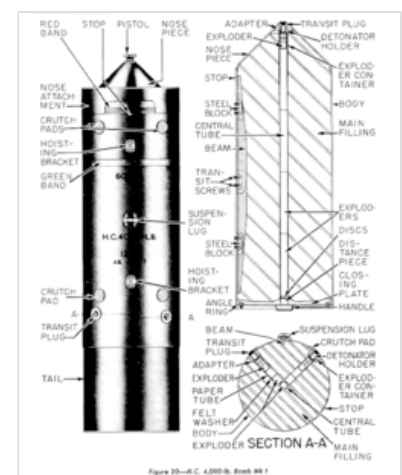
Examples of military nomenclatures around the world using the term mine shell or equivalent:

-  – English *High-Explosive, High-Capacity (HEHC)*^{[1][3][2]}
-  – French: *obus à mine* and *obus fougasse* for land based munition^[7] According to older French weapons standardization, the term "à mine" was used for landmines buried deeper than 3 m (9.8 ft) in the ground, while the term "fougasse" was used for landmines buried less than 3 m (9.8 ft) below the ground (see *Fougasse (weapon)*).^[6] This indicates that there probably were differences in effect between 'Obus á mine' and 'Obus fougasse' shells.
-  – German: *Minengeschoß (M-gesch)* for conventional mine shells or *Minenbrandgranate (M-brgr)* for incendiary mine shells^[9]
-  – Hungarian: *rombológránát*^[10]
-  – Italian: *granata mina*^[11]
-  – Norwegian: *minegranat*^[12]
-  – Spanish: *granada mina*^[13]
-  – Swedish: *mingranat (mingr or mgr)*^[14]

Other applications

The damage effect of mine shells, often described as "pressure wave damage" or simply named "mine damage/mine effect", has historically been applied to other types of ordnance than cannon shells, most notably aerial bombs (example names: German: *Minenbombe*,^[15] English: *High-Capacity bomb*).

Air-to-air rockets configured with mine-shell equivalent warheads have also at times featured "mine" in their name. For example, the German WW2 rocket R4M was an abbreviation of *Rakete, 4 Kg, Minenkopf*, meaning "Rocket, 4 kg (8.8 lb), Mine-head" in English.^[16] Interestingly, mine-rockets developed in Sweden after WW2 were named *sprängraketer* (high-explosive rockets), even though their initial design was directly based on the German R4M.^[17]



British WWII 4,000 lb
(1,810 kg) "high-capacity" (HC)
bomb Mk.1

History

Dedicated mine shells originate in Europe. The origin of the name "mine shell" is unknown but Italy was using the name by 1884 (Italian: *Granata Mina*).^[11]

Mine shells were used in a lot of different types of high caliber cannons, howitzers, and mortars on both land and on water around the turn of the century, before seeing a decline after World War I. (See the article Minenwerfer for the use of mine shells in mortars during World War I.) The use of rebar-reinforced fortifications during World War I probably made the shell obsolete as its effect on fortifications was negated by the rebar.



Imperial German 25 cm (9.8 in) "mine-mortar" (German: *Minenwerfer*) during World War I.

German use of mine shells in World War II

During World War II, mine shells would see a resurgence as the Germans started to use the type in small caliber (initially 20 mm) automatic weapons, both to arm the Luftwaffe's fighter aircraft and for Flak. This was an innovation, as prior to this, mine shells had only been constructed in large calibers for technical reasons. Larger shells were usually produced by casting, smaller calibers by drilling the cavity for fuse and explosives into a solid steel shot, and neither process was effective at making small projectiles with walls that were sufficiently thin yet strong enough to work as a mine shell. While small thick-walled shells fired from automatic guns performed well against ground targets, they were more limited in anti-aircraft use.

Development

In the late 1930s, the Germans began to pay attention to these shortcomings during the trials of the 20 mm MG FF cannon.^[8] Its conventional high-explosive rounds were judged unsatisfactory in the anti-aircraft role, for the reasons mentioned above. As a result of these trials, the German air ministry, the *Reichsluftfahrtministerium*, or RLM for short, ordered the development of mine shells for the 20 mm MG FF cannon in 1937.^[8] To make such shells in 20 mm (0.787 in) caliber, German ordnance engineers had to try new methods of construction; what they came up with was a round made from high quality drawing steel, manufactured in the same way in which cartridge cases are made.^[18] These new 20 mm mine shells were first used against the RAF in 1940, and proved highly successful. Even when the British and later, to a limited extent the Americans equipped their fighters with autocannon, they always used conventional ammunition. The difference in payloads between these rounds and the Luftwaffe's mine shells was significant. Considering the high-explosive rounds alone as an example: the 20 mm mine shells used in MG FF/M cannons (and later in the MG 151/20) both had a 17 g (0.60 oz) HE filling while British and American autocannon shells of the same calibre, but markedly heavier, could carry only 10–12 g (0.35–0.42 oz), while the typical filler load in the conventional 20 mm shells of the original MG-FF was a mere 4.5–6.5 g (0.16–0.23 oz).

As mentioned above, one problem with the new ammunition was that due to their lightweight nature, the new 20 mm mine shells produced insufficient recoil to operate the 20 mm MG FF cannon. This required a modification of the recoil mechanism so the cannon could fire this new shell, but this in turn made it unsafe to fire the old, conventional rounds. In an effort to avoid the chambering of incorrect ammunition, the modified weapon was redesignated the 20 mm MG FF/M, *M* for *Minengeschoß*.

Deployment

Germany first used Minengeschoß ammunition during the Battle of Britain when MG FF/M armed Bf 109E's and Bf 110C's flew missions over from mainland Europe to Britain. Although the shells themselves proved deadly, the guns had a poor rate of fire, relatively sluggish muzzle velocity and an inadequate magazine ammunition feed, and were soon to be replaced by the belt-fed MG 151. This new type was originally introduced as a Minengeschoß-firing heavy machine gun, in 15 mm; but then it was realised that the earlier cannon-sized mine shells were more effective, and so a new larger cartridge (20x82mm) was created for the weapon. The adapted gun, (more precisely designated the MG 151/20), became the Luftwaffe's standard 20 mm autocannon until the end of the war,^[19] and with its high fire rate coupled with good ballistics and high-explosives payload for its caliber was overall among the best aircraft armament of the conflict.^{[20][21]}



A Messerschmitt Bf 109E-4, the first Bf 109 deployed with the 20 mm MG FF/M "mine shell capable" cannon.



German 30x90mmRB mm rounds as used by the MK 108 cannon. The sectioned round is a mine shell (*Minengeschoß*).

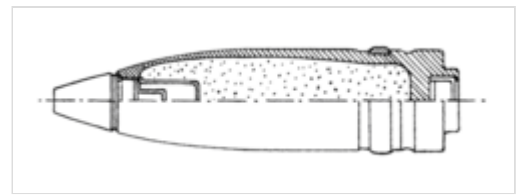
As the possibilities of this new application for mine shells became better understood, the Luftwaffe found they had created a potential game-changer as the recoil/velocity ratio made it possible to create larger caliber guns that would have low enough recoil to be effectively mounted on conventional aircraft, while at the same time achieving useful velocities. Moreover, as the volume of a cylinder is proportional to the square of its radius, and as cannon shells tend to the shape of a cylinder, the Minengeschoß design when applied to larger calibers allowed a dramatic increase in explosive payload and power. One such weapon was the 30 mm (1.18 in) MK 108

which became highly militarily significant during the second half of the war, when the Allies began to mount their enormous bombing onslaught on German cities. So large was the increase in internal volume indeed that it proved worthwhile to the Germans to refine these projectiles by making them more streamlined, sacrificing a little of this capacity, but thus partly compensating for the lower momentum characteristic of the Minengeschoß design. These streamlined mine shells for the 30 mm MK 108 were designated Ausf.C.^[22] and featured 72 grams (2.54 ounces) of nitropenta (PETN), compared to the original blunt-nosed Ausf.A which had 85 grams (3.00 ounces) of PETN. (*Note that the Ausf.B was a training shell without explosives.*) See below for a comparison with modern ammunition loads.

Mine shells were also adopted for use in ground attack cannons like the high-velocity 30 mm MK 103, among others,^{[23][24]} as well as anti-aircraft guns like the 2 cm Flak 30/38, 3.7 cm Flak 18/36/37/43 and 5 cm FlaK 41.^[10]

Further development

At the end of the WWII the Germans started to develop several autocannons in caliber 55 mm (2.17 in) for use in aircraft and anti-aircraft guns against allied aircraft. Some examples being the Gerät 58^[25] anti-aircraft gun and the MK 112 and MK 115 aircraft cannons. The reason the Germans settled on an unconventional caliber of 55 mm was due to it being the smallest caliber that could shoot down a heavy bomber with a single mine shell.^[10] Calculations and tests had revealed that a single explosion of 400 grams (0.882 pounds) of PETN or RDX explosive mixes could effectively take out a heavy bomber in a single hit.^[10] Due to requirements for minimalism, it was calculated that a caliber of at least 55 mm (2.17 in) was needed to deliver this load. Interestingly Germany already manufactured mine shells in caliber 50 mm (1.97 in) for guns like the 5 cm FlaK 41, but these only had an explosive charge capacity of 360 grams (0.794 pounds).^[10]



German 55 mm mine shell design from WWII.

In the end, no 55 mm mine shells seems to have been deployed by Germany during World War II. The allied bombing of Germany delayed weapons-research and made production of the 55 mm weapons impossible.

However, the Germans did manage to deploy a 55 mm air-to-air rocket named "Rakete, 4 Kg, Minenkopf" (Rocket, 4 kg, Mine-head), or R4M for short, at the end of 1944.^[16] It was fitted with a 55 mm "high-capacity", or "mine" warhead filled with 520 gram of the explosive-mixture "HTA 41"^[16] (also known as "HTA 15"),^[26] which consists of 40% Hexogen (RDX), 45% TNT and 15% aluminium.^[26] The shell walls of the warhead were only 0.8 mm (0.0315 in) thick.^[16] These rockets were fitted to several German aircraft at the end of the war, most notably the Me 262, which used them to great effect during their limited service life.^[16] On one occasion, Me 262s armed with R4M rockets shot down 25 B-17 bombers out of a group of 425 within a very short time without any losses.^[16]



B-24 Liberator bomber shot down by a 55 mm R4M mine-warhead air-to-air rocket.

Post-war use

After the defeat of Germany in World War II, several countries started using mine shells for their own post-war aircraft and anti-aircraft armament, for example the "high-explosive" shells of Britain's ADEN cannon and the French DEFA 540 were effectively clones of German wartime equivalents.^[27] The guns themselves were developments of the German Mausier MK 213.^[27]



British 30 mm ADEN cannon, a further developed copy of the German Mauser MG 213, was specifically developed to fire mine shells.^[27]



French 30 mm DEFA cannon, a further-developed copy of the German Mauser MG 213, was specifically developed to fire mine shells.^[27]



Swedish 30 mm mine shells in 30 × 111 mm J for use in the 30 mm ADEN Mk3 cannon.^[27]

Left: Swedish production

Right: British production

Even in comparison to modern designs, some of the WW2 payloads quoted above are impressive as not even the PGU-13/B HEI round for the GAU-8/A Avenger gun of the A-10 Warthog or the 30 mm OFZ shell of Russian GSh-30-1 and GSh-30-6 cannons come close to the German WW2 mine shells of the same caliber - 72–85 g (2.54–3.00 oz) compared to 58 g (2.05 oz) and 48.5 g (1.71 oz) respectively for the PGU-13B and OFZ.

Sweden having experience with the shell type from earlier developed several different mine shells in several different calibers after the war. Some examples being a mine shell variant for the 20 x 110 Hispano cartridge^[28] and one for the 57 x 230R Bofors cartridge.^[29]

The type is still used today in autocannons such as the Mauser BK-27^[30] but there is no known use of the type as it was originally used.

Weapons adapted with mine shells

Aircraft weapons

- 20 mm MG FF/M cannon
- 20 mm MG 151 cannon
- 20 mm Mauser MG 213
- 20 mm Hispano-Suiza HS.404
- 27 mm Mauser BK-27
- 30 mm MK 101 cannon

- 30 mm MK 103 cannon
- 30 mm MK 108 cannon
- 30 mm Mauser MK 213
- 30 mm ADEN cannon
- 30 mm DEFA cannon
- 30 mm Oerlikon KCA
- 37 mm BK 3,7
- 50 mm Rheinmetall BK-5
- 50 mm MK 214A cannon

Anti-air weapons

- 2 cm Flak 30, Flak 38 and Flakvierling 38
- 3.7 cm Flak 18/36/37
- 3.7 cm Flak 43
- 5 cm FlaK 41
- R4M

Artillery weapons

- 7.58 cm Minenwerfer
- 9.15 cm leichtes Minenwerfer System Lanz
- 17 cm mittlerer Minenwerfer
- 22.5 cm Minenwerfer M 15
- 24 cm schwerer LadungsWerfer Ehrhardt
- 24 cm schwere Flügelminenwerfer IKO
- 24 cm schwere FlügelminenWerfer Albrecht
- 240 mm Trench Mortar
- 9.45-inch Heavy Mortar
- 25 cm schwerer Minenwerfer
- 28 cm howitzer L/10

Notes

- British term can also be found written in reverse: *high-capacity, high-explosive (HCHE)*.

See also

- High-explosive incendiary
- High-explosive incendiary/armor-piercing ammunition
- Raufoss Mk 211

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 - [Luftwaffe Aircraft Gun and Cannon Ammunition of WW II \(http://www.inert-ord.net/luft02h/index.html\)](http://www.inert-ord.net/luft02h/index.html)
 - [Description of Rheinmetall-Borsig MK 108 30 mm Cannon and ammunition \(http://www.luft46.com/armament/mk108.html\)](http://www.luft46.com/armament/mk108.html)
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